Edge Federated Learning Study

Abstract. This thesis studies federated learning when participating edge devices differ widely in compute, memory and data distribution. We propose an adaptive client selection scheme combined with per-layer differential privacy noise, and show that it reduces communication rounds while bounding privacy loss on non-IID benchmarks.

Federated optimisation on phones and gateways suffers from stragglers whose local epochs finish long after the aggregation deadline. Our scheduler estimates each device budget from recent round latencies and assigns proportional workloads.

Gaussian noise calibrated per layer keeps the privacy accountant within epsilon of four while losing under two points of accuracy on FEMNIST and Shakespeare. Ablations isolate the contribution of momentum correction at the server.

A deployment across forty Raspberry Pi nodes in the campus laboratory confirms the simulated speedups and exposes energy constraints that motivate the final chapter on battery-aware participation.